Camouflaged Objects Cannot Hide – We Can See Through and Localize

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Abstract. Deep learning (DL) models for complex vision tasks, including camouflaged object detection (COD) and medical image analysis (MIA), are computationally expensive, making their deployment difficult in resource-constrained environments (e.g., edge devices, and surveillance systems). Although image cropping or downsampling to reduce extraneous image area is feasible for generic images, it remains a challenge for COD or MIA tasks due to their low signal-to-noise ratios. To address this issue, we propose a preprocessing block that frames the localization of camouflaged objects as a conjunction search task that aggregates texture, frequency, and convexity features. Subsequently, a uniform bounding box drawn from the aggregated saliency localizes the camouflaged object and separates it from the croppable extraneous image areas, improving the quality of data. As the efficacy study shows, the proposed method reduces DL training time, energy consumption and CO_2 emission by roughly 30-50% at a cost of minimal impact on accuracy – typically seeing about 1-3% drop, as tested across multiple datasets and models. The proposed preprocessing-focused efficiency gain is equally applicable to other computer vision tasks and may help make SOTA models deployable for resource-constrained environments.

1 Introduction

Deep Learning (DL) models are the de facto standard for computer vision tasks [28, 52], spanning object detection [67], image classification [55], camouflaged object detection [16], and medical image analysis [40]. These tasks heavily rely on convolutional neural networks (CNNs) and Transformers, which are known to be data and computational resource intensive [31, 64]. For instance, performing inference on a single 224×224 image can require approximately 1.1 billion multiplications [54]. For the deeper networks often required for COD and MIA tasks, the problem is exacerbated. The use of large, high-resolution images further complicates the situation [31, 44], affecting the deployment of DL models in edge devices for many applications (e.g., remote sensing and MIA) [64]. Although efforts have been made to mitigate this issue through different model reduction techniques, such as pruning, precision reduction, and sparsity, they remain inadequate for strict energy budgets [23, 72].

Among alternatives, image downsampling eases memory and computation cost, and is commonly used [44, 49, 56, 64]. However, eliminating image pixels for COD [16, 34, 37] and MIA [51, 69] tasks may eliminate discernible boundary information, making object localization significantly more challenging for both CNN and Transformer DNNs. An alternative approach is to reduce trainable image area. However, it remains inadequate for COD, as attributing high saliency to object region, as adopted in standard saliency detection approaches, is inappropriate [19] and can be ineffective in object centroid estimation (Fig. 2).

Precisely, in COD, camouflaging relates inversely to the objects saliency. Because camouflaged objects foil feature search (color, edge, texture, shape, etc.), performed in the human vision and perception system [59], leveraging multiple information channels, such as the conjunction search, becomes crucial as suggested in the feature integration theory [10, 59, 62].

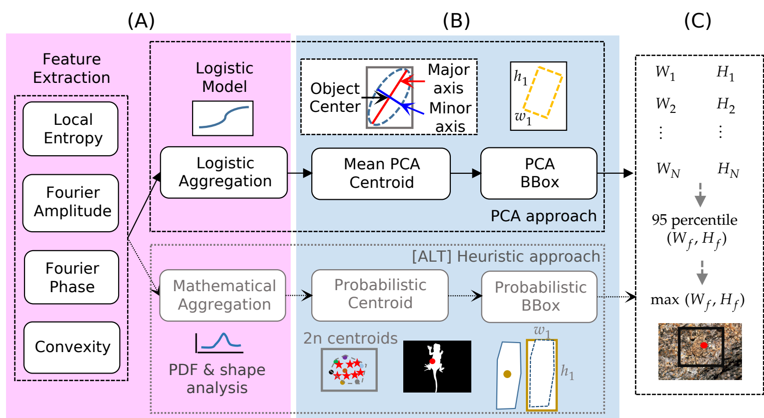

Fig. 1: Proposed methods — probabilistic or heuristic approach (dimmed) and principal component analysis (PCA) based approach (highlighted) — that integrate texture, Fourier and convexity analysis for uniform bounding box generation in camouflage and medical images. Each method consists of three sub parts: A) Camouflaged signal extraction and amplification B) Centroid estimation and intermediate bounding box drawing, and C) Statistical uniform bounding box drawing to crop extraneous regions.

As a solution to the camouflaged object localization problem, we devise a conjunction search that integrates frequency-domain features, geometry, and image texture, facilitating removal of extraneous image area. Precisely, we localize the camouflaged object through integration of the mentioned features and subsequently crop the image using a statistically obtained uniform bounding box, separating salient and extraneous regions. We propose two separate methods for generating bounding boxes – probabilistic and principal component analysis (PCA) based – where the PCA-based approach is a simpler and much faster solution (See Table 3). Experimental results show that the proposed PCA image cropping mechanism retains 90% of the object area in approximately 90-95% of images across various vision tasks. Moreover, extraneous area removal reduces training cost by about 30-50% while maintaining competitive accuracy (reducing roughly $1-3\%$ in most cases) across different vision tasks, including COD [16,22] and MIA [13, 17, 51, 65]. Keeping the energy budget in mind, we also adapt the SINet for CPU-only execution [16] Our specific contributions are as follows:

- Identification and integration of features (entropy, Fourier, and convexity) capable of localizing the camouflaged object, using two distinct methods – probabilistic and PCA.
- Extensive validation of the proposed methods, by using them as a preprocessing block with the SOTA DL models, on different datasets. Data shows, proposed method reduces computational expense by about 30-50% at a cost of only $1-3\%$ accuracy, across SOTA DL models.
- Design of a CPU-only pipeline using SINet [15] and U-Net that, along with the preprocessing block, performs competitively against GPU systems.

2 Related works

Automatic image cropping, frequently used in thumbnail formation in photography, localizes and captures salient image regions [9,18,63,66]. Saliency detection in generic images often rely on pixel-level characteristics: intensity, color, texture and luminance [36], but, when pixel level characteristics are insufficient, frequency domain information – amplitude [24] and phase [41] – can be leveraged for better performance. However, traditional saliency detection approaches become ineffective for COD or MIA, as objects downplay discerning saliency cues through countershading [58], disruptive coloration and edges [45, 53], and compromised luminance. In addition, context dependency in camouflage breaking further complicates the object identification process [39]. For example, a gradient based approach that effectively breaks camouflage in a variety of images, struggles in images where camouflage is attributed to countershading [58]. Therefore, to tackle camouflaged objects in any context and ensure their effective localization, multiple information cues need to be incorporated, making conjunction search crucial [10, 59, 62].

Currently, deep learning models have emerged as the standard for camouflaged object localization and segmentation tasks. To deal with camouflage, some CNN-based approaches relied on feature extraction that combined context, texture, and frequency information to identify objects immersed in the background [38, 68, 71]. In a multitask learning framework, CNN-based models achieved significant improvements in COD through classification [30], edge detection [34], reconstruction [70], and object ranking [34]. Among many CNN-based models, FEDER [22] decomposes image features into frequency bands and exploits the more informative band, while the edge reconstruction model addresses the edge disruption problem. A data augmentation module CamoFA [29], swaps frequency components between input image and generated reference images, to enhance the accuracy of models. Another model, ZoomNeXt [43] incorporates zooming in and out, while simultaneously addressing texture loss for COD [47]

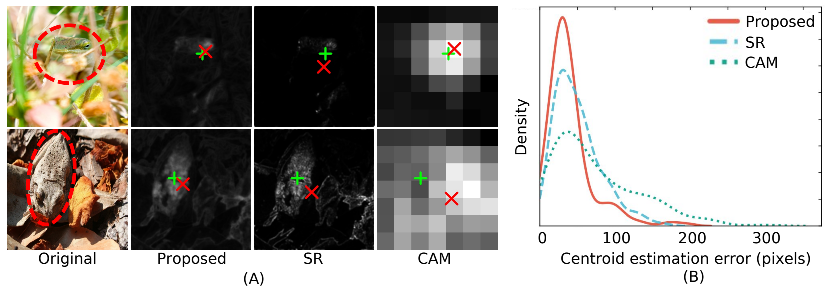

Fig. 2: (A) Inadequacy of the existing off the shelf localization approaches – for instance Spectral Residual (SR) and Class Activation Maps (CAM) – in estimating the object centroid in COD images. The figure illustrates the estimated centroid (red $\times$) is the closest to the true center of the object (green $+$) for the proposed method, where the other methods estimate a centroid far away or outside the actual object. (B) A probability distribution plot illustrating the distribution of centroid estimation errors (in pixels). The distributions result from 200 COD10K images selected based on target location, where half of the images have centroid distance $\leq$ 85 pixels from center of image and the other half have centroid distance $>$ 85 pixels from center.

and accumulation of structural information into single-scale, often overlooked in other CNN-based approaches [32, 33].

However, these CNN-based SOTA models have their limitations. For instance, SINet-v2 [15] and ZoomNet [42] have limitations in the extraction of global features and incorrect prediction of object regions. Interestingly, vision transformer-based [14] approach overcomes the limitations through capturing long-range feature dependencies, but it suffers from the inefficiency of accumulating local texture information, required for camouflage breaking [26]. To tackle the shortcoming of vision transformers in COD, the Feature Shrinkage Pyramid Network (FSPNet) [26] enhances the adjacent features to addresses the locality exploration. Regardless of their respective strengths and weaknesses, most DNN models rely on computationally expensive devices, limiting their deployability in edge devices and resource constrained environments.

3 Proposed method

Visual attention is fundamentally stochastic, so the accuracy of predicting an object's centroid from its saliency map is inherently limited. The object localization process requires detecting abrupt intensity changes, and camouflaging traits further aggravate detectability. In the presence of such uncertainty, probabilistic detection of abrupt change, for instance, through change-point detection, mimics human attention and accounts for context-dependency and distributional shifts induced by noise [3, 27]. Accommodating each saliency cue to localize objects via change-point assessment is one of two approaches proposed for predicting objects' centroids and subsequent uniform box drawing. Interestingly, the change-point-based probabilistic approach applies not only to a single image but also to video processing and is adaptive to ambient changes. In an alternative approach, we extract the most informative regions from the aggregated saliency using PCA; however, we use only the first two components, ignoring low-variance components. That is, centroid prediction from the most informative saliency region via the PCA constitutes the other method devised in this paper. Irrespective of whether it's probabilistic or PCA-based, the approaches devised comprise three common steps: i) Extraction and amplification of camouflaged signal, ii) Capture of amplified signal using intermediate image specific bounding boxes, iii) Attainment of dataset specific uniform bounding box, from the intermediate image specific bounding boxes, via statistical analysis.

3.1 Feature cues for Conjunctional search

Features cues for Conjunctional search resulted from approximately 200 different combinations of methods. Frequency domain exploration involved shape based filtering, phase magnitude swapping, energy spectral density, and cumulative sum profiling. Different textural investigation included Haralick features [21], local binary patterns, local variance, and entropy based region isolation via peak, change-point, and histogram analysis. Further exploration involved gradient extraction, Gaussian highpass filters, computational geometry (Voronoi tessellation, Delaunay triangulation, convex hulls) [12], FREAK keypoints [2], non-linear attentions, and classical detectors (e.g. Canny [7] and watershed [5]). The combination comprising Local Shannon-entropy, Fourier, and convexity information outperforms other combinations tried as in Table 4.

Local Shannon-entropy Camouflaged objects exhibit local textural variations around their edges or in parts of their bodies. Local entropy can be used to capture these granular textural variations in a small neighborhood [8,46,50,73]. The local entropy, denoted as $\mathcal{H}(I_G(x,y))$, of a gray scale image $I_G(x,y)$ is defined over a window Ω of dimension $K \times K$ as Eq. 1:

$$\mathcal{H}(I_G(x,y)) = \sum_{g=0}^{L-1} p_g \log_2 \frac{1}{p_g} = \sum_{g=0}^{L-1} \frac{n_g}{K^2} \log_2 \left(\frac{K^2}{n_g} \right) \tag{1}$$

However, local entropy may not always be effective for COD and MIA images, as objects successfully eliminate or minimize local variation through different camouflaging strategies [16,22].

Fourier domain information Fourier-domain information is effective for breaking camouflage [68]. Following [1], the phase-only transform (PHOT) filters the Fourier-phase and eliminates the regular or background patterns. Additionally, we apply log transform on the amplitude spectrum, followed by Gaussian filtering, to suppress background noise and highlight the camouflaged object. For a grayscale image $I_G(x, y)$ with dimension $W \times H$, where $x = 0, 1, \ldots W-1$ and $y = 0, 1, \ldots H-1$, the 2D Discrete Fourier transform is denoted as $\mathcal{F}(u, v)$:

$$\mathcal{F}(u,v) = \sum_{x=0}^{W-1} \sum_{y=0}^{H-1} I_G(x,y) e^{-j2\pi ux/W} e^{-j2\pi vy/H} = |\mathcal{F}(u,v)| e^{j\phi(u,v)} \tag{2}$$

where $|\mathcal{F}(u, v)|$ and $\phi(u, v)$ denote amplitude and phase spectrum, respectively. In PHOT, we use Eq. 2 to calculate the Fourier transform $\mathcal{F}(u, v)$ and normalize it by the magnitude,

$$\text{PHOT}(u,v) = \mathcal{F}(u,v)/\mathcal{M}(u,v); \qquad S_{\text{PHOT}}(x,y) = \mathcal{F}^{-1}(\text{PHOT}(u,v)) \tag{3}$$

In addition, we apply a log transform on the amplitude spectrum followed by a 2D Gaussian filter G_γ in the frequency domain to obtain $S_{\text{amp}}(x, y)$, where γ is the filter's standard deviation.

$$S_{\text{amp}}(x,y) = \mathcal{F}^{-1}\left(\left(\log(1 + |\mathcal{F}(u,v)|) * G_\gamma(u,v) \right) e^{j\phi(u,v)} \right) \tag{4}$$

Convexity assessment of objects: $\boldsymbol{D}_{\textbf{arg}}$ Camouflaged objects can contain sub-regions of convex (or concave) connected parts [57], which can be leveraged in their localization process. In order to extract the convex regions from an image, the proposed method deploys the differentiation of argument (D_{arg}) operator [57, 58]. For an input image $I_G(x, y)$, the gradient direction $C\theta(x, y)$ quantifies change in x and y direction, and is calculated as Eq. 5:

$$\nabla I_G(x,y) = \left(\tfrac{\partial}{\partial x} I_G(x,y), \tfrac{\partial}{\partial y} I_G(x,y) \right); C\theta(x,y) = \arctan\left(\frac{\frac{\partial}{\partial y} I_G(x,y)}{\frac{\partial}{\partial x} I_G(x,y)} \right)$$

The derivative of $C\theta(x, y)$ in the y direction results in the operator CY_{arg} (Eq. 5), which is not isotropic. To infuse isotropy, the image is rotated $0, 90, 180, 270$ degrees, and sum of CY_{arg} of the rotated images is calculated for the final D_{arg}.

$$CY_{\text{arg}} = \tfrac{\partial}{\partial y} C\theta(x,y), \quad D_{\text{arg}} = \textstyle\sum (CY_{\text{arg}})|_{\{0,90,180,270\}} \tag{5}$$

It should be noted that, performance of the D_{arg} deteriorates for iamges containing convex and rough background.

3.2 Feature aggregation

Non learning-based aggregation In order to integrate the saliency maps for the probabilistic method, we utilized a non learning-based aggregation. The aggregated saliency value $S_{agr}(x, y)$ can be obtained from m saliency maps $\{S_k | 1 \leq k \leq m\}$ computed from an image I_G, using the Eq. 6 from [6],

$$S_{agr}(x,y) = \frac{1}{Z} \sum_{k=1}^{m} \zeta(S_k(x,y)); \zeta(s) = \exp(s) \tag{6}$$

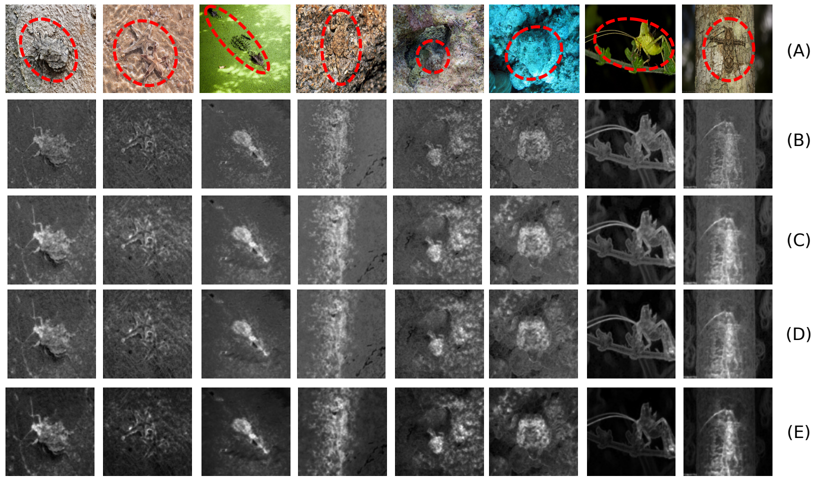

Fig. 3: (A) Original images from COD10k dataset with objects marked using dashed circle. (B) Aggregated saliency maps obtained through addition of saliency channels (C) Aggregate maps calculated through addition, but the final map has been normalized by dividing each pixel by the sum of all pixels. (D) Aggregated maps through addition and normalization, where contrast of individual saliency maps have been amplified by taking the exponential of each pixel. (E) Aggregation using a logistic regression model.

In Eq. 6, $1/Z$ is a normalizing factor, where Z equals the sum of the amplified saliency values across all pixels, guaranteeing the final map to be a valid probability density function (PDF). On the other hand, the ζ function is used to amplify the contrast of the individual saliency maps. Aggregation without the normalization and zeta parameter has been used in our proposed probabilistic method and the results are presented in Fig. 3B. Successively, results obtained after adding the normalization parameter is presented in Fig. 3C, and Fig. 3D shows the results of saliency aggregation keeping all terms of Eq. 6.

Learning-based aggregation The proposed PCA approach utilizes a learning based aggregation strategy – logistic regression – to obtain robust aggregation. In order to train the logistic model from [35], we construct a feature vector $\mathbf{f}(x, y)$ per pixel (x, y) from the training set, $\mathbf{f}(x, y) = (S_1(x, y), \ldots, S_m(x, y))$, where S_k is the K-th saliency map. The ground truth mask from the train set is used as the labels (Y) for the feature vectors. After training the logistic model, we obtain m weights $(\omega_1, \omega_2, \ldots, \omega_m)$ and a bias term (ω_{m+1}), using which we compute the weighted sum for feature vectors of the test set images.

$$S_{agr}(x, y) = \sigma_{\text{sig}} \left(\sum_{k=1}^{m} \omega_k S_k(x, y) + \omega_{m+1} \right), \quad \text{where} \quad \sigma_{\text{sig}}(s) = \frac{1}{1 + \exp(-s)}. \tag{7}$$

Table 1: Algorithmic formalization of the proposed bounding-box drawing approaches. The left panel outlines the Principal Component Analysis (PCA)-based methodology utilizing logistic regression, while the right panel details the Probability density-centric approach utilizing surrogate centroids.

Algorithm 1 PCA-based Approach

Require: Set of N images, m Channels $S_k \in \{\mathcal{H}(I_G), D_{\arg}, S_{\mathrm{PHOT}}, S_{\mathrm{AMP}}\}$
Ensure: Uniform cropping dimension $\max(H_f, W_f)$
for each image $i \in \{1, \ldots, N\}$ **do**
  Calculate aggregated saliency via logistic regression:
  $S_{agr}(x, y) \leftarrow \sigma_{\mathrm{sig}}\left(\sum_{k=1}^{m} \omega_k S_k(x, y) + \omega_{m+1}\right)$
  Filter S_{agr} to retain significant pixels (x, y)
  Compute spatial centroid $\mathbf{c}$ (mean vector)
  Perform PCA to extract eigenvalues λ_1, λ_2 and eigenvectors $\mathbf{v}_1, \mathbf{v}_2$
  $a \leftarrow \sqrt{\lambda_1}$, $b \leftarrow \sqrt{\lambda_2}$ {Half-lengths}
  Calculate 2σ bounding vertices $\mathbf{p}_{1..4}$:
  $\mathbf{p}_{1,2} \leftarrow \mathbf{c} + 2a\mathbf{v}_1 \pm 2b\mathbf{v}_2$
  $\mathbf{p}_{3,4} \leftarrow \mathbf{c} - 2a\mathbf{v}_1 \pm 2b\mathbf{v}_2$
  Compute intermediate dimensions (W_i, H_i) from $\mathbf{p}_{1..4}$
end for
$W_f \leftarrow$ 95-percentile$(W_{1\ldots N})$
$H_f \leftarrow$ 95-percentile$(H_{1\ldots N})$
return $\max(H_f, W_f)$

Algorithm 2 Probabilistic Approach

Require: Set of N images, m Channels $S_k \in \{\mathcal{H}(I_G), D_{\arg}, S_{\mathrm{PHOT}}, S_{\mathrm{AMP}}\}$
Ensure: Uniform cropping dimension $\max(H_f, W_f)$
for each image $i \in \{1, \ldots, N\}$ **do**
  for each channel $k \in \{1, \ldots, m\}$ **do**
    Apply morphology and contour analysis to S_k
    Apply change-point detection to obtain binary map $S_k^{(B)}$
    Draw Convex Hull to extract centroid C_k and dimensions (w_k, h_k)
  end for
  Merge $S_k^{(B)}$ using merging function $M(\cdot)$
  Form area A via convex hull over $\{C_1, \ldots, C_m\}$
  Generate N_{sc} surrogate centroids (SC) on A via spatial Poisson process:
  $P\{\mathbb{N} = n\} = \frac{\exp(-\rho A)(\rho A)^n}{n!}$
  Compute final centroid (See supplement)
  Compute intermediate dimensions:
  $W_i \leftarrow \mathrm{median}(w_1, \ldots, w_m)$
  $H_i \leftarrow \mathrm{median}(h_1, \ldots, h_m)$
end for
$W_f \leftarrow$ 95-percentile$(W_{1\ldots N})$
$H_f \leftarrow$ 95-percentile$(H_{1\ldots N})$
return $\max(H_f, W_f)$

3.3 Signal capture using intermediate bounding boxes

PCA-based approach To extract the geometric distribution properties of the aggregate saliency region $S_{agr} \in \mathbb{R}^{W \times H}$, obtained from learning based aggregation logistic regression, a two-component PCA is fitted to the filtered set of spatial coordinates. This analysis yields the spatial centroid of the region, denoted as the mean vector $\mathbf{c}$, alongside the principal axes of variance. The orientation and dimensional extents of the bounding box are defined by the eigenvalues (λ_1, λ_2) and their corresponding eigenvectors $(\mathbf{v}_1, \mathbf{v}_2)$ derived from the covariance matrix of the spatial coordinates. The half-lengths of the bounding box along its principal axes are determined by the square root of the explained variances, denoted as $a = \sqrt{\lambda_1}$ and $b = \sqrt{\lambda_2}$. The square root of the explained variances represents the standard deviations (σ) of the pixel distribution along the principal axes. To ensure comprehensive enclosure of the target object, the half-lengths of the bounding box, denoted as a and b, are scaled to establish a 2σ boundary, capturing approximately 95% of the spatial variance.

Probabilistic approach Unlike the PCA method, the probabilistic approach treats each saliency channel independently, acknowledging that individual channels may underperform on certain images. To mitigate these fluctuations, it introduces surrogate centroids generated via a Poisson process. First, each saliency channel S_k undergoes morphological, contour, and change-point analysis (see supplement) to produce binary images, which are then enclosed by convex hulls. The centroid $C_1, \ldots, C_m$ and minimum enclosing rectangle dimensions (w_k, h_k) of each hull are recorded.To approximate the final centroid, a new convex hull of area A is formed from the initial centroids $C_1, \ldots, C_m$. Within A, we generate N_{sc} uncorrelated surrogate centroids, $SC_1, \ldots, SC_{N_{sc}}$, using a homogeneous spatial Poisson point process:

$$P\{\mathbb{N} = n\} = \frac{\exp(-\rho A)(\rho A)^n}{n!} \tag{8}$$

with $E[\mathbb{N}] = \rho A$. Averaging these N_{sc} surrogate centroids effectively absorbs spatial perturbations, yielding a highly stable final centroid (Fig. 1, dimmed). Algorithm 2 in Tab. 1 summarizes this bounding box generation process.

3.4 Uniform cropping dimension

With centroid fixation and an intermediate bounding rectangle achieved through the process outlined in Fig. 1, Tab. 1 and the previous sections, we consider it for the projected uniform bounding box used for image cropping. The uniform cropping dimension is measured using only the training set of images. For the probabilistic approach, we take the median of the candidate heights (h_k) and widths (w_k) across the different feature channels k to obtain the intermediate rectangle dimensions (W_i, H_i) for a given image i. This computation continues for all $i \in \{1, \ldots, N\}$, where N is the total number of images in the dataset. For the PCA-based approach (Fig. 1, highlighted), we directly extract (W_i, H_i) from the intermediate rectangles obtained through PCA, as in Tab. 1 Algorithm 1. Finally, for both approaches, we calculate the 95-percentile of all intermediate widths $W_{1 \ldots N}$ and heights $H_{1 \ldots N}$, to obtain the global dimensions W_f and H_f. We select $\max(H_f, W_f)$ as the final uniform bounding-box dimension.

3.5 CPU-only architecture design

Our design philosophy exploits parallelism at the thread, process, and instruction levels to maximize resource utilization across the available logical and physical cores (see Table 5a). We implemented cache-blocking techniques that partition large tensors into smaller tiles that fit within the L2 cache (2MB per P-core). We use 256×256 pixel tiles that align with the cache line size (64 bytes) to minimize cache misses and memory stalls for convolution operations. We assign the performance cores to primary mathematical operations, including matrix multiplications and convolutions. These cores deliver high performance for compute-intensive tasks. Efficiency cores are dedicated to secondary operations: activation

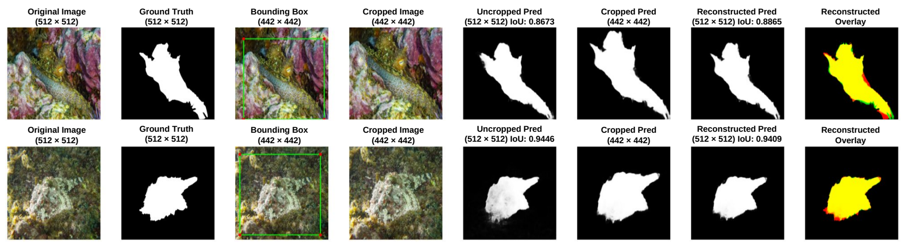

Fig. 4: Visual comparison of segmentation performance. For each example (row), we compare the baseline *Uncropped Prediction* with our *Reconstructed Prediction*. The final overlay visualizes the performance: Yellow (Overlap), Red (Ground Truth only), and Green (Prediction only). For details of the reconstruction process see supplement.

functions, normalization layers, and utility tasks. We reserved at a minimum 20% of total cores for system processes and coordination tasks, preventing interference with the main training workload. A sample implementation of U-Net and its performance is outlined in Table 5a (see supplemental information).

3.6 Experimental setup

Datasets We evaluated our approach on three COD datasets and two medical image datasets. The COD benchmarks include: COD10K [16], NC4K [34], and CAMO [30] . For medical evaluation, we train with the Polyp dataset (CVC-Colon [4]) and the Skin_Lesion dataset (ISIC 2016 [20]). **Models:** We fine-tune and evaluate our performance against a range of architectures, from traditional convolutional baselines to modern Transformer-based models. These include: (1) U-Net [48], (2) SINet [16] with a ResNet-50 backbone, (3) ZoomNeXt [43] (RN50), and the PVT-v2-B2 [61] based variants (4) HitNet [25] and (5) EndoKED [60]. **Metrics:** Segmentation performance is compared using Intersection Over Union (IoU), Precision (Pre.), Recall(Rec.), F1, Structural Similarity Index (SSIM) and Mean Absolute Error (MAE). Computational efficiency is reported as per-image total training time (TT, minutes), energy consumption (E, Wh), and CO2-equivalent emissions (g). Metrics marked with ↓ are better when lower; all others are better when higher. The measurements of training time, energy and CO_2 emission does not include the cost of proposed PCA based cropping, since it is negligible (See Table 3).

4 Results

4.1 GT area preservation and background area reduction

Similarity between object and background, renders conventional centroid estimation approaches ineffective in COD (see Fig. 2), resulting in inaccurate cropping. Table 6 quantifies cropping performance of the proposed PCA-based approach

Table 2: Evaluation of uncropped (U) vs. PCA approach based cropped and reconstructed (C+R) methods. Results show that C+R provides significant computational benefits with minimal accuracy gaps; e.g., **ZoomNeXt on COD10K** improves **IoU by 3%** while reducing **TT by 32.6%**. All cropped images are reconstructed to original dimensions for fair comparison. TT: Training Time; E: Energy Consumption; CO_2: Emissions. (↓) indicates lower is better; no arrow indicates higher is better.

Data	Model	Method	Performance						Efficiency		
			IoU (%)	Pre. (%)	Rec. (%)	F1 (%)	SSIM (%)	MAE↓	TT (m)↓	E (Wh)↓	CO_2 (g)↓
COD10K	HitNet	U	75.3	84.6	87.2	85.9	93.8	0.026	64.2	16.2	6.8
		C+R	72.8	86.9	81.8	84.3	93.5	0.028	34.0	8.7	3.6
	SINet	U	64.2	80.2	76.5	78.3	80.1	0.048	33.1	8.3	3.5
		C+R	61.4	79.7	72.8	76.1	79.1	0.051	19.3	4.8	2.0
	ZoomNeXt	U	64.2	78.7	77.7	78.2	87.9	0.041	72.9	18.3	7.6
		C+R	67.2	85.2	76.1	80.4	91.3	0.035	49.1	12.4	5.2
NC4K	HitNet	U	81.1	88.5	90.6	89.5	93.4	0.031	75.5	19.0	7.9
		C+R	78.1	87.5	88.0	87.7	92.6	0.036	61.5	15.5	6.5
	SINet	U	72.2	84.3	83.5	83.9	81.5	0.055	39.8	10.0	4.2
		C+R	68.0	80.8	81.1	81.0	78.0	0.065	20.2	5.1	2.1
	ZoomNeXt	U	71.7	82.0	85.0	83.5	86.6	0.051	100.2	25.2	10.5
		C+R	74.4	88.7	82.3	85.3	88.9	0.043	68.5	12.3	5.1
CAMO	HitNet	U	71.4	85.0	81.7	83.3	79.0	0.063	25.3	6.4	2.7
		C+R	70.0	84.0	80.7	82.3	56.1	0.074	13.7	3.4	1.4
	SINet	U	55.3	72.2	70.2	71.2	62.3	0.119	13.0	3.3	1.4
		C+R	55.9	74.5	69.1	71.7	60.9	0.117	8.0	2.0	0.9
	ZoomNeXt	U	59.1	76.5	72.2	74.3	77.7	0.095	33.2	8.5	3.5
		C+R	63.8	83.4	73.1	77.9	84.1	0.078	21.0	5.3	2.2
Polyp	EndoKED	U	85.6	93.8	90.8	92.3	95.3	0.014	6.2	1.6	0.7
		C+R	83.2	90.8	90.8	90.8	92.1	0.023	3.3	0.8	0.3
	U-Net	U	82.7	88.1	93.0	90.5	10.1	0.120	6.2	1.6	0.7
		C+R	74.5	90.9	80.6	85.4	10.0	0.119	3.4	0.9	0.4
Skin Les.	EndoKED	U	87.0	92.4	93.6	93.0	77.8	0.049	21.4	5.4	2.3
		C+R	86.2	93.3	91.8	92.6	86.2	0.047	10.7	2.9	1.2
	U-Net	U	85.6	93.2	91.3	92.2	84.8	0.050	21.1	5.3	2.2
		C+R	84.2	93.1	89.8	91.4	73.5	0.056	12.0	3.0	1.3

across five datasets. In every dataset, at least 70% of the GT object area is preserved in virtually all images (99% of images on COD10K, NC4K, CVC-colon, and ISIC; 100% on CAMO). The 80% retention threshold is met by over 95% of images on every dataset. The stricter 90% threshold is achieved in approximately 90% of images across the COD benchmarks and over 94% on the medical datasets. Even at the 95% retention level, between 75% and 90% of images are correctly enclosed, depending on dataset complexity. Concurrently, the uniform bounding box reduces total image area by 19–28%, corresponding to dimension reductions of 10–15% per axis. Importantly, the CVC-Colon dataset achieves the highest area reduction (28%) at full 95% retention.

4.2 Location independent localization

The assumption of the object being situated at the center of the image is crucial to many vision related tasks, which proves ineffective in the context of camouflaged and medical image analysis. Fig. 2A visually demonstrates the effectiveness of proposed PCA method in locating the objects, even when they are off

Table 3: Computational efficiency comparison between the probabilistic and PCA-based methods. Results are reported as mean ± standard deviation. The PCA-based approach demonstrates a substantial reduction in processing time per image.

Method	Time per Image (s)	Peak Memory (MB)
Probabilistic	10.33 ± 1.50	31.02 ± 0.09
PCA-based	0.26 ± 0.03	33.94 ± 0.37

Table 4: Effectiveness of different approaches on the COD10K test set: GHP: Gaussian High-Pass Filter, LE: Local Entropy, A_{RC}: Attention (root cube), A_{EXP}: Attention (exponential), OB: Otsu Binarization, D: Dilation applied, FM: Fourier Magnitude Phase Extraction, FP: Fourier Phase-Only Reconstruction (Regularity Removal), IR: Image Reconstruction, F: Fourier, P: Phot, Darg: Darg, E: Entropy, LR: Logistic Regression.

No.	Method	DSC	SSIM	No.	Method	DSC	SSIM
1	GHP → LE → A_{RC} → OB → D	0.28	0.60	10	FP → GHP → LE → OB	0.27	0.55
2	GHP → LE → A_{RC} → OB	0.27	0.66	11	FP → A_{RC} → GHP → LE → OB	0.27	0.49
3	GHP → LE → A_{EXP} → OB → D	0.28	0.60	12	FP → A_{EXP} → GHP → LE → OB	0.27	0.55
4	GHP → LE → A_{EXP} → OB	0.27	0.66	13	FP → GHP → LE → OB	0.27	0.55
5	GHP → LE → OB → D	0.28	0.60	14	FM → GHP → A_{RC} → IR → LE → OB → D	0.25	0.59
6	GHP → LE → OB	0.27	0.66	15	FM → GHP → A_{RC} → IR → LE → OB	0.25	0.68
7	FM → GHP → IR → LE → OB	0.25	0.43	16	FM → GHP → IR → LE → OB → D	0.19	0.25
8	FP → GHP → A_{RC} → LE → OB	0.27	0.50	17	FM → GHP → IR → LE → OB	0.20	0.40
9	FP → GHP → A_{EXP} → LE → OB	0.27	0.55	**18**	**F → P → $\mathbf{D}_{arg}$→ E → LR**	**0.28**	**0.70**

center. Moreover, Fig. 2B quantitatively demonstrates localization performance of the proposed method along with others, for images in two categories: 1) close to the center of the image, and 2) far from the center of the image. The proposed method shows a tight clustering around low centroid estimation error. On the other hand, SR and CAM exhibit a significantly wider distribution, highlighting the higher centroid estimation accuracy of the proposed method in location independent localization tasks.

4.3 Computational cost reduction and accuracy

To measure the computational gain from the reduced trainable area, we fine-tuned the models on their respective benchmark datasets, as detailed in Section 3.6. On several model-dataset pairs the cropped plus reconstructed (C+R) variant achieves equal or superior accuracy to the uncropped baselines (Tab. 2), when PCA based approach is used for cropping due to its efficient (See Tab. 3) and robust localization performance. We attribute this performance to the high signal to noise ratio in the training images that reduce burden on the model to learn background representations. The most prominent example is ZoomNeXt on COD10K, where IoU improves from 64.2 to 67.2 (+4.7%), F1 from 78.2 to 80.4 (+2.8%), and SSIM from 87.9 to 91.3 (+3.9%). Comparable gains are observed for ZoomNeXt on CAMO and NC4K. For SINet and HitNet on COD10K and NC4K, accuracy parity is largely maintained: F1 differences are within $1-3\%$ in most cases. The average normalized performance of PCA method across datasets

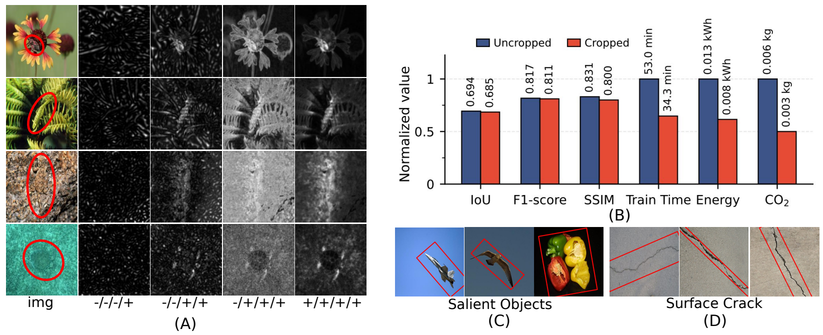

Fig. 5: A) The iterative improvements obtained through addition of the proposed channels. Fourier/Entropy/Phot/D_{arg}, with + and - denote inclusion and exclusion, respectively. (B) Normalized computational metrics, highlighting significantly reduction in computation and emissions with only minimal accuracy reduction. (C-D) Usability of proposed method in ROI localization on salient objects and concrete crack.

Table 5: Hardware computational analysis: **(a)** Impact of multicore CPU architecture on training time. **(b)** Performance comparison between GPU and CPU implementations; the CPU-only approach requires roughly 4.1× the training time of the GPU.

(a) CPU Core Utilization (U-Net / CVC-Colon)

CPU Cores	Train Epochs	Train Time (s)	Val. Dice
8	15	1924.4	0.6718
16	15	1429.5	0.6665
32	15	1254.0	0.6875

(b) GPU vs. CPU Comparison (SINet / COD10k)

GPU		CPU	
Train Time (m)	Test F1	Train Time (m)	Test F1
11.7042	0.7156	48.0785	0.6981

Table 6: Evaluation of the proposed cropping method on different datasets (original image: 512×512). The number of images retaining specific percentages of the ground truth (GT) area, alongside the corresponding dimension and area reduction achieved.

Dataset (Total Images)	Images Ret. GT Area (≥)				Reduction Metrics		
	70%	80%	90%	95%	Crop. Dim.	Dim. Reduc.	Area Reduc.
COD10k (2025)	2021	1993	1892	1738	442×442	14%	25%
NC4k (1119)	1116	1110	1057	966	452×452	12%	22%
Camo (250)	250	248	228	197	460×460	10%	19%
CVC-colon (112)	112	112	106	94	434×434	15%	28%
ISIC (379)	379	379	362	338	462×462	10%	19%

is demonstrated in 5B. Visually, Fig. 4 confirms this minimal accuracy trade-off, demonstrating a strong overlap between the original ground truth and predictions from models trained on cropped images. The probabilistic approach also substantially reduces area while retaining accuracy (see supplement).

4.4 CPU only architecture performance

The unavailability of powerful GPUs for training deep learning models is a significant obstacle, which can be alleviated by training models only on CPU. To this end, a CPU only architecture was proposed and tested on a limited scale with 1000 COD10k samples, using the SINet model. The results presented in Tab. 5b compare the GPU implementation of the SINet model with the CPU implementation, in terms of training time and F1 score. From the Tab. 5b, it is visible that the training time of the CPU implementation grew slightly over four times, while the F1 score is comparably close to the GPU implementation, proving the usefulness of the architecture in environments with resource constraint.

4.5 Ablation study

To assess the contribution of individual information cues in the overall efficacy of the proposed system, we performed a detailed ablation study using the COD10K dataset. The qualitative results of the ablation study is presented in 5A, which demonstrates iterative improvement with the addition of information channels – Entropy, Fourier, Darg and Phot. The original images with marked camouflaged objects are in the leftmost column, and the inclusion or exclusion of a specific channel is mentioned in the labels using plus and minus.

The quantitative results presented in Tab. 7 outlines removal of Entropy, Fourier or Darg channels directly contribute to reduced area retention in cropped images. Among these channels, removal of Entropy and Darg have greater impact on area retention performance, meaning stronger contribution. Interestingly, removal of the phase channel (Phot) increases area retention, but at the cost of lower area reduction in cropped images, making it suboptimal. The best results in terms of both GT area retention and background information reduction is obtained when every channel is included in the proposed method.

5 Discussion

While we achieve significant computing and energy gain, object decimation by the bounding box occasionally misses portions of object areas and could be further improved. Our continuing works assess directionality extraction, mean shift algorithm [11] and conditional random field [35] for further improvements.

Table 7: Ablation study of different channel combinations and their effects on GT area retention, image area reduction and cropped dimension. (Original image:512×512).

Channel Names				Performance Metrics		
Entropy	**Fourier**	**Phot**	**Darg**	**Area Ret.(≥90%)**	**Area Ret.(≥95%)**	**CroppedDim.**
	✓	✓	✓	131	113	370
✓		✓	✓	178	162	422
✓	✓		✓	196	188	474
✓	✓	✓		171	148	412
✓	✓	✓	✓	187	177	458

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Supplementary Material for: Camouflaged Objects Cannot Hide – We Can See Through and Localize

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1 Camouflaged object localization

1.1 PCA Based Approach

The probabilistic algorithm is useful as it does not require any dataset specific learning or extensive parameter tuning, but it relies heavily on heuristics. To overcome the shortcomings of the probabilistic method we proposed the principal component analysis (PCA) based approach, which uses logistic regression for saliency aggregation and PCA for centroid as well as rectangle arm length estimation. Specifically, the PCA based approach leverages the entropy, convexity and fourier saliency maps to calculate a combined saliency map using logistic regression and then uses PCA to find an optimal cropping window. In this approach, the mean of the PCA of the saliency map serves as the centroid (c) and the two unit vectors (v_1 and v_2) along with standard deviations (a and b) in the direction of principal and secondary axis are used to obtain four coordinates of the non uniform rectangle (p_1-p_4).

$$
\begin{aligned}
p_1 &= c + a \cdot v_1 + b \cdot v_2 \\
p_2 &= c + a \cdot v_1 - b \cdot v_2 \\
p_3 &= c - a \cdot v_1 - b \cdot v_2 \\
p_4 &= c - a \cdot v_1 + b \cdot v_2
\end{aligned}
$$

The per image rectangle is then used to obtain a final statistical uniform bounding box in the same process described in Fig. 2 lower part.

1.2 Probabilistic Approach

The probabilistic approach achieves automatic image cropping through applying a series of post processing steps on saliency maps obtained from different channels. Precisely, the saliency maps from the texture, convexity, Fourier channels as well as their aggregation (Fig. 1B) go through multitude of analysis: probability density function analysis, shape analysis, contour analysis etc.; before obtaining a cleaned binary image which serves as seeds for centroid calculation (Fig. 1A). The exact steps to obtain the cleaned images are described in Alg. 1, 5, 3, 4, 2. The cleaned binary maps are then used to statistically choose the uniform bounding box dimension, as in Fig. 2.

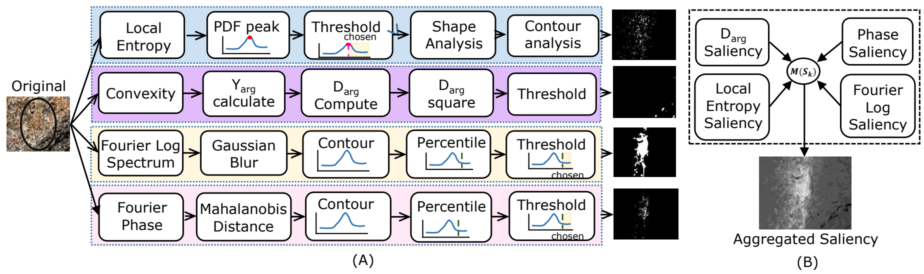

Fig. 1: (A) Pipeline for probabilistic approach to generate seeds for centroid calculation (B) Aggregation of saliency for centroid calculation using merging function $M(S_k)$

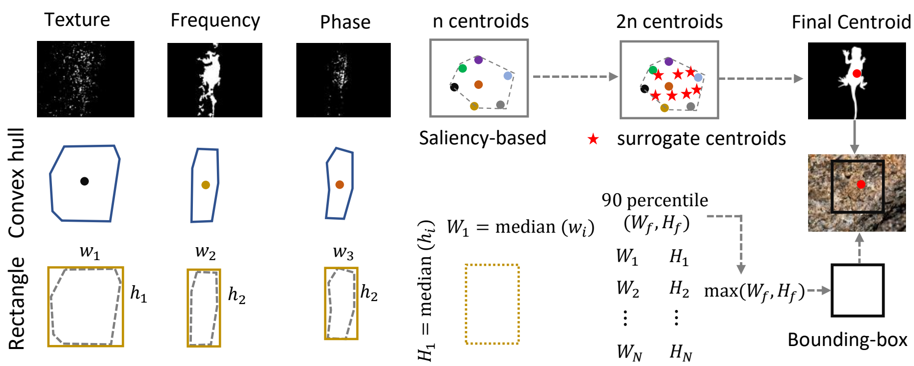

Fig. 2: Process of incorporating multi cue information to obtain centroid and uniform bounding box.

2 Uniform Bounding Box Drawing:

Most deep learning (DL) model architectures are designed to operate on images of uniform dimension, imposing a constraint for any processing module meant to reduce the image size. The formulation of a uniform bounding box over all the images, where the saliency region used for bounding box drawing can take different shapes and sizes, is particularly challenging. To tackle this problem, we introduce a statistical approach in order to find the box, which can be used to crop the image while keeping the object of interest intact within the frame.

In both of the proposed methods — probabilistic and PCA — we use the same statistical process to obtain the box. The process of choosing the final uniform bounding box along with centroid estimation using the probabilistic approach is explained in Fig. 2. We first calculated the image specific non-uniform bounding box, which is the median of minimum enclosing rectangles over the convex hulls of the saliency maps, and later used the arm length of these non-uniform boxes to determine the size of the final uniform box. Specifically, we captured the non-uniform bounding box arm length per image and then later used a percentile threshold on the arm length data to obtain a final bounding box dimension. The threshold is placed in order to accommodate the larger objects while eradicating the outliers, meaning maximum retention of object area and maximum reduction of background area. For the PCA based approach the rectangles in Fig. 2 is obtained through calculation of the major axis and minor axis using principal component analysis of the aggregated saliency map (elaborated further in Section 03).

Method	Dim. (Test)	Train Time (m)	Pre.	Rec.	F_1	$F_{0.5}$	F_2	Energy (kW)	CO_2 (kg)
Dataset: COD10K (1000 Samples) – Model: SINet [2]									
Uncropped (Original)	512×512	11.70	0.8324	0.6297	0.7157	0.7821	0.6619	0.0029	0.0012
Cropped + Recon.	426×426	6.86	0.7367	0.6947	0.7143	0.7279	0.7028	0.0017	0.0007
Dataset: CVC-Colon (380 Samples) – Model: EndoKED [14]									
Uncropped (Original)	512×512	5.22	0.9156	0.8743	0.8944	0.9070	0.8823	0.0013	0.0006
Cropped + Recon.	466×466	3.74	0.8865	0.8587	0.8721	0.8808	0.8641	0.0009	0.0004
Dataset: MSRA10k (1000 Samples) – Model: F^3Net [15]									
Uncropped (Original)	512×512	14.01	0.8986	0.8874	0.8930	0.8963	0.8896	0.0035	0.0015
Cropped + Recon.	426×426	8.18	0.9042	0.8449	0.8735	0.8917	0.8561	0.0021	0.0009
Dataset: ISIC-2016 (379 Samples) – Model: EndoKED [14]									
Uncropped (Original)	512×512	5.21	0.8848	0.9458	0.9243	0.8964	0.9329	0.0013	0.0006
Cropped + Recon.	466×466	3.76	0.9007	0.8919	0.8962	0.8989	0.8936	0.0009	0.0004
Dataset: COD10K (1000 Samples) – Model: SINet [2] – Down-sampled to 20%									
Uncropped (Original)	409×409	10.99	0.8288	0.6245	0.7123	0.7783	0.6581	0.0032	0.0013
Cropped + Recon.	341×341	5.90	0.7042	0.6944	0.6992	0.7032	0.7002	0.0014	0.0014

Table 1: Evaluation comparing cropped images with uncropped (original) images on accuracy and computational metrics using deep convolutional neural network models. For a fair evaluation, the cropped images were reconstructed to the ground truth image dimension before evaluation. All reported metrics are the average of 3 separate experiments, with each experiment run for 30 epochs. The gray-highlighted row shows that images can be further down-sampled to get additional computational gain without compromising much accuracy, a technique that is also applicable to the cropped images.

3 Analysis of Probabilistic Cropping Performance

Table 1 demonstrates the efficacy of the proposed probabilistic cropping method, highlighting a highly favorable trade-off between computational efficiency and model accuracy across multiple datasets and architectures. **Computational and Environmental Efficiency:** The most significant advantage of the probabilistic cropping method is the drastic reduction in computational overhead. By effectively localizing the target object and discarding extraneous background data, the input dimensions are notably reduced (e.g., from 512×512 to 426×426 for COD10K). This dimensional reduction translates to substantial accelerations in training time. For instance, when utilizing SINet on the COD10K dataset, the training time decreases from 11.70 minutes to 6.86 minutes—an approximate

Algorithm 1 Channel Entropy Processing

Require: Grayscale image I_{gray}
Ensure: Binary entropy map $E^{(B)}$
1: $E_{\text{ent}} \leftarrow \text{Entropy}(I_{\text{gray}}, \mathcal{K}^{5\times5}_{\text{disk}})$
2: $H_{\text{peak}} \leftarrow$ bin with maximum frequency in Histogram(E_{ent})
3: $E_{\text{ent}}[E_{\text{ent}} < H_{\text{peak}}] \leftarrow 0$ {Threshold below peak}
4: $\mathcal{C} \leftarrow \text{ExtractContours}(\text{MorphologicalGradient}(E_{\text{ent}}))$
5: $x_1, x_2 \leftarrow$ change-points of area PDF in $\mathcal{C}$
6: $\tau_{\text{area}} \leftarrow \begin{cases} x_1, & \text{single change point} \\ \frac{1}{2}(x_1 + x_2), & \text{average of two points} \\ \frac{1}{2}\left(x_1 + \frac{x_1+x_2}{2}\right), & \text{combined average} \end{cases}$
7: **return** ContourFiltering($\mathcal{C}, \tau_{\text{area}}$)

Algorithm 2 Channel Fourier Processing

Require: Grayscale image I_{gray}
Ensure: Binary Fourier map $F^{(B)}$
1: $\mathcal{M}, \mathcal{P} \leftarrow \text{FourierTransform}(I_{\text{gray}})$
2: $F_{\text{gauss}} \leftarrow \text{GaussianBlur}(\text{IDFT}(20 \log_{10} \mathcal{M}, \mathcal{P}))$
3: $\tau_{\text{int}} \leftarrow 90^{\text{th}}$ percentile of F_{gauss}
4: $F_{\text{bin}} \leftarrow (F_{\text{gauss}} > \tau_{\text{int}}) \times 255$ {Binarize}
5: $\mathcal{C} \leftarrow \text{ExtractContours}(F_{\text{bin}})$
6: $\tau_{\text{area}} \leftarrow 90^{\text{th}}$ percentile of areas in $\mathcal{C}$
7: **return** ContourFiltering($\mathcal{C}, \tau_{\text{area}}$)

41% reduction. Crucially, this optimization directly mitigates the environmental impact of model training, yielding proportional decreases in both energy consumption (kW) and CO_2 emissions (kg) across all tested models, including EndoKED and F^3Net. **Accuracy Retention:** Despite the aggressive removal of spatial data, the probabilistic method successfully preserves the integrity of the salient features, resulting in minimal accuracy degradation. Across the medical imaging datasets (CVC-Colon and ISIC-2016), the F_1 scores experience only marginal drops (e.g., a decrease of just 0.022 for CVC-Colon) while cutting training time by nearly 30%. Interestingly, in complex scenarios like the COD10K dataset, while precision slightly decreases, the method actually improves recall

Algorithm 3 Channel Phot Processing

Require: Grayscale image I_{gray}
Ensure: Binary phot map $P^{(B)}$
1: $P_{\text{gauss}} \leftarrow \text{GaussianBlur}(\text{CalculatePhot}(I_{\text{gray}}))$
2: $\tau_{\text{int}} \leftarrow 90^{\text{th}}$ percentile of P_{gauss}
3: $P_{\text{bin}} \leftarrow (P_{\text{gauss}} > \tau_{\text{int}}) \times 255$ {Binarize}
4: $\mathcal{C} \leftarrow \text{ExtractContours}(P_{\text{bin}})$
5: $\tau_{\text{area}} \leftarrow 90^{\text{th}}$ percentile of areas in $\mathcal{C}$
6: **return** ContourFiltering($\mathcal{C}, \tau_{\text{area}}$)

Algorithm 4 Contour Filtering

Require: Set of contours $\mathcal{C}$, area threshold τ_{area}
Ensure: Filtered contour set $\mathcal{C}_{\text{filt}}$
1: **return** $\{C \in \mathcal{C} \mid \text{Area}(C) \geq \tau_{\text{area}}\}$

Algorithm 5 Channel Convexity (D_{arq}) Processing

Require: Grayscale image I_{gray}
Ensure: Binary convexity map $D^{(B)}$
1: $D_{\text{norm}} \leftarrow \text{Normalize}(\text{CalculateConvexity}(I_{\text{gray}})^2, [0, 255])$
2: $D^{(B)} \leftarrow (D_{\text{norm}} > 0.7 \times 255) \times 255$ {Threshold at 178.5}
3: **return** $D^{(B)}$

(from 0.6297 to 0.6947), indicating that the cropped bounding boxes effectively force the network to focus on the camouflaged object, maintaining a highly stable F_1 score of 0.7143 compared to the uncropped baseline of 0.71

4 Object localization using non-uniform boxes

Training and testing state-of-the-art (SOTA) deep learning (DL) models require uniform image dimensions. To satisfy this constraint, the proposed method uses statistical analysis to generate uniform bounding boxes. Although useful for model training, statistically chosen uniform bounding boxes inevitably introduce extraneous background data into many images. For applications lacking rigid image dimension restrictions, the proposed methods can localize objects using a minimum enclosing non-uniform box. The proposed methods generate these non-uniform bounding boxes via two distinct approaches. Probabilistic method employs the rotating calipers algorithm with probabilistic saliency maps and PCA method combines PCA with aggregated saliency derived through logistic regression. Fig. 3B demonstrates the proposed method's effectiveness in localizing objects using non-uniform bounding boxes generated by the rotating calipers algorithm. As the figure shows, the generated bounding boxes capture almost the entire object area, with few exceptions (e.g., the bounding box excludes the crocodile's tail in Fig. 3B, top-left).

5 Information Cues for Conjunction Search

The proposed conjunction [13, 16] search comprises feature integration through different information cues forming their respective saliency map. The three information cues that we apply in the conjunction search for object localization are

- Image texture features through Shannon local entropy that considers a window of $w \times w$ dimension and captures local variation.

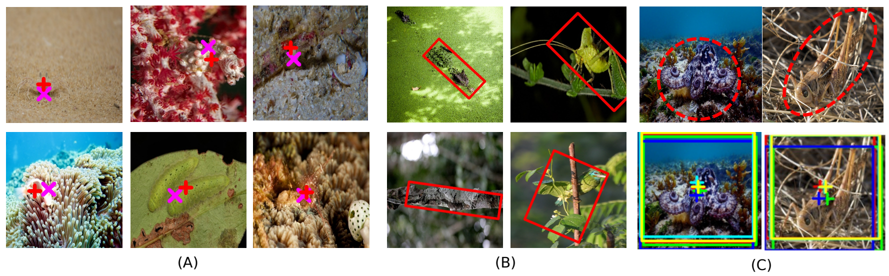

Fig. 3: (A) Performance of the proposed probabilistic method in localizing objects, independent of their location in the image. The cross (magenta) is the approximate centroid and plus (red) is the actual (B) Efficacy of the probabilistic method in drawing non uniform minimum bounding boxes on camouflaged objects (C) Demonstration of the contribution of information cues: ensemble of all methods (blue), entropy (green), Fourier (cyan), phot (yellow) and D_{arg} (red) are projected on the same image

– Frequency domain features through transforming the image using Discrete Fourier Transform (DFT). Two alternative methods, focusing on amplitude spectra and phase of the image, respectively, form two different information cues.
– Convexity assessment of the object of interest assuming that target objects, for instance polyp or other living objects, which are of convex shape.

We performed an extensive analysis to identify cues that provide relevant object information assessing the feature variation in a locality. Precisely, texture synthesis, as performed, include Gray Level Co-occurrence Matrix (GLCM) [5], Shanon local entropy analysis [11] and the Gabor filter [4,8] approach to identify the texture variation in images across different datasets.

6 Texture Synthesis

6.1 Gray Level Co-occurrence Matrix (GLCM)

Among the many features extractable from the GLCM-based image texture analysis, a few that we calculate and compare with for alternative datasets are contrast, dissimilarity, homogeneity, ASM, energy and correlation. For the calculation of these features, the formulation, as outlined in [5], is used.

The accuracy of texture analysis depends on the size of the window W, for which the target localization accuracy varies considerably. Using the DSC metrics [17], we find that smaller window sizes more accurately locate the target object, as quantified by DSC against the ground truth provided. An earlier study supports neither a too small W nor a too large W in calculating local entropy [3]. We also found this to be optimal for local entropy analysis. To summarize, the steps performed in the GLCM-based analysis include:-

- All input images were resized to a 256x256 size for the GLCM-based analysis to speed up the computation for an explorative study over alternative datasets.
- To capture the texture variation in a local neighborhood, a sliding/gliding window of size $w \times w$, where $w \in \{5, 7, 11\}$ is passed over the entire image to generate six texture images capturing energy, homogeneity, correlation, dissimilarity, contrast, and angular second moment separately.
- In GLCM-based texture analysis, we calculated the values against 0 degree only, 45 degree only, 90 degree only, and 135 degree only, as well as taking the mean from all four angles and the maximum from all four angles to assess the extractable information among the different texture features tried.

6.2 Gabor Filter Analysis

Gabor filters [4, 8] mimic the human vision system and have been a standard feature extractor in image analysis and vision tasks. Here, texture synthesis also considers the Gabor derivative in both x and y directions, with angular quantization for rotations of $0, 45, 90, 135$ degrees. The exact mathematical expression for 2-dimensional Gabor function for texture analysis of Image is:

$$\begin{aligned} &G(x, y; \omega, \theta, \sigma_x, \sigma_y) \\ &= \frac{1}{2\pi\sigma_x\sigma_y} \exp\left(-\frac{1}{2}\left(\frac{x^2}{\sigma_x^2} + \frac{y^2}{\sigma_y^2}\right) + j\omega\left(x\cos\theta + y\sin\theta\right)\right) \end{aligned} \tag{1}$$

where σ is the spread, θ is the orientation, and the term ω denotes the frequency. Using a Gabor kernel, upon 2-D Gabor filtering we capture the Gabor gradient along X and Y for texture analysis for an image.

6.3 Local Entropy

The statistical disorder in a neighborhood, as measured by local entropy analysis, is also considered a means of differentiating whether an object is hidden in a particular region. Precisely, we calculated the local entropy using the Shannon entropy definition [11] with different window dimensions and found that a window size of 5×5 was the optimal choice across various images and alternative datasets.

6.4 Local Entropy is more informative than GLCM textures

The Haralick texture or GLCM features can identify informative regions in many images [6]. However, the effectiveness in camouflaged setting is limited or insufficient. The visual examples in medical camouflage polyp in Figure 4 shows, Haralick features such as homogeneity, dissimilarity and contrast is unable to match the local entropy saliency map, which attributes higher intensity to polyp regions. For COD10k objects a similar trend follows where local entropy strongly highlights the object region and the Haralick features, though have some traces of the object, produce a saliency map too weak to be taken into consideration (See Figure 6). Addditionally, a quantitative analysis on COD10K 100 images in Table 2 shows that local entropy based centroid using off the shelf approaches – OTSU binarization and moment calculation – yields the lowest mean, median and maximum error 23.58, 22.51 and 55.76 pixels respectively, indicating more informative region capture by local entropy. Even though the GLCM features fail to identify regions of interest in camouflaged setting, it captures useful image regions in more salient disease images, that is skin lesion. In Figure 5 it is visible that apart from the Angular Second Moment (ASM) Haralick feature all the other features capture the region of interest, but they are not as strong and discriminative as the local entropy for the shown cases. Interestingly, in some of the cases the region of interest is actually darker and the background is brighter, establishing an inverse relationship not seen in other image sets. Such behavior of Haralick features make it even more difficult to use it in a versatile solution.

Table 2: Centroid Error Distance (px) Summary across GLCM texture methods and local entropy. All metrics are calculated over $N = 100$ samples per method.

Method	**Mean (↓)**	**Std (↓)**	**Median (↓)**	**Min (↓)**	**Max (↓)**
ASM	30.89	29.09	24.15	1.00	158.90
Contrast	25.37	19.24	23.59	2.00	132.31
Correlation	25.81	14.23	24.68	1.00	70.94
Dissimilarity	24.11	13.81	22.52	3.16	63.32
Energy	30.58	24.46	25.29	1.00	155.57
Homogeneity	37.29	25.65	31.98	2.24	144.81
Local Entropy	23.58	13.01	22.51	4.12	55.76

7 Image Binarization

The area of a binarized image region depends on the exact threshold used, making the saliency map sensitive to the thresholding mechanism. The global and adaptive thresholds have limitations, lacking robustness or being computationally intensive. Situation worsens further for the intended application comprising images of multifaceted feature variations such as composition and appearances (normal images vs. COD, MIA), contrast, luminance, color, composition, and intensity of pixels and subregion. Considering such a diverse image pool different vision tasks possess, and the underlying uncertainty of the intensity function of various features across various tasks, we histogram-to-PDF (f) conversion of a grey-scale feature image FI to binarize it or during the subsequent processing of the binarized image. As an example, the binarized (B) salient region from local entropy (E) feature image, $E(B)$, initially is produced using the peak $f^{\max}$ of

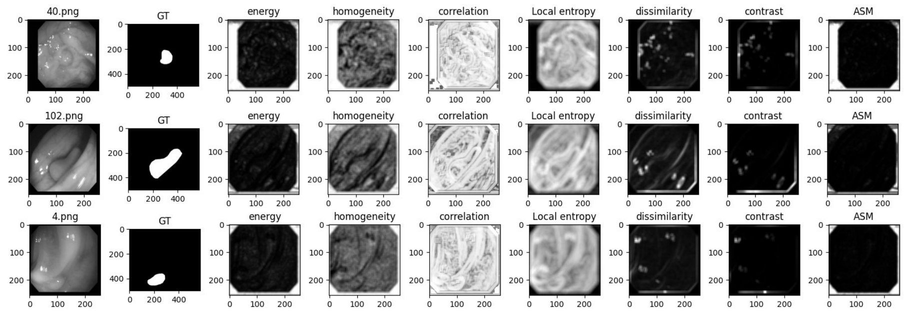

Fig. 4: GLCM polyp: Different Gray Level Co-occurrence Matrices (GLCM) features for the CVC-colon (polyp) dataset. The images visually show the insufficiency of different GLCM features, in providing meaningful information about the object of interest in the polyp dataset. The local entropy channels are also presented to show visual comparison in between the GLCM features and local entropy. Local entropy clearly provides more informative images as seen in the figure.

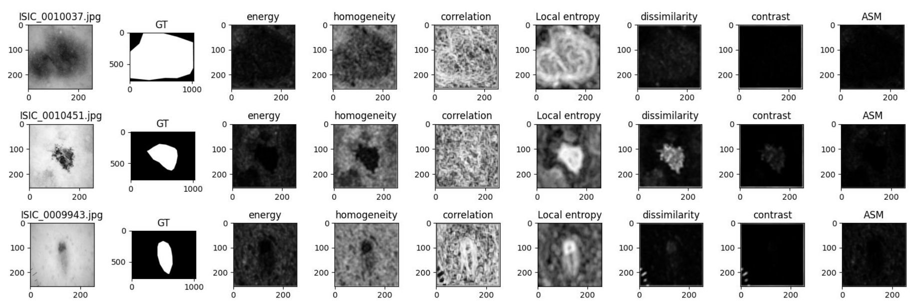

Fig. 5: GLCM Skin lesion: Different GLCM features for the ISBI (skin lesion) dataset. The images visually show the ineffectiveness of different GLCM features in providing meaningful information about the object of interest in the polyp dataset. The local entropy channels is also presented to show visual comparison in between the GLCM features and local entropy. Local entropy clearly provides more informative images.

the PDF f at α and extract the preliminary candidate pixel position for the saliency map. Here, the PDF f of local entropy grayscale image $E(G)$ is generated through histogram analysis followed by Kernel Density Estimaton (KDE) process.

$$f^{\max} = \max\,[f_0, \ldots f_{255}] = f_\alpha, f^{\text{trunc}} = [f_\alpha \ldots f_{255}]$$
$$E(B)_{\text{trunc}(i,j)} = \begin{cases} 0 \equiv \mathit{black}, & \text{if } f(i,j) < f_\alpha \\ 255 \equiv \text{white}, & \text{otherwise} \end{cases} \tag{2}$$

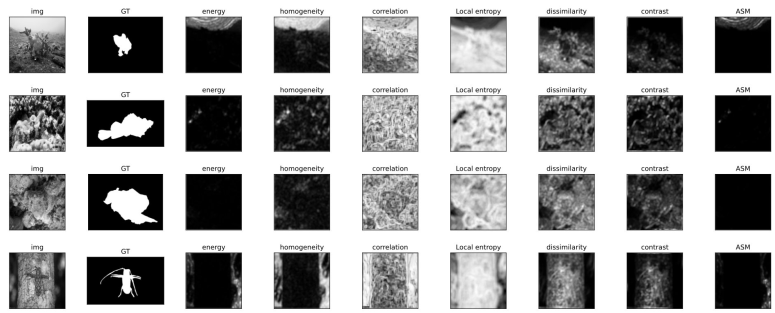

Fig. 6: GLCM COD10k: GLCM features and local entropy for COD10k images. The local entropy in each image highlights the object area partially or completely, while the GLCM features fail to reliably highlight the object. In the cases where GLCM highlights traces of the object successfully, the local entropy provides a stronger and more discriminative result.

The $E(B)_{\text{trunc}}$ undergoes a morphological gradient analysis followed by erosion with a diamond kernel of size 3. The modified $E(B)_{\text{trunc}}$ undergoes a contour (C) analysis, which we use to obtain PDF, denoted as $f_C(E)$ of the area of each contour found.

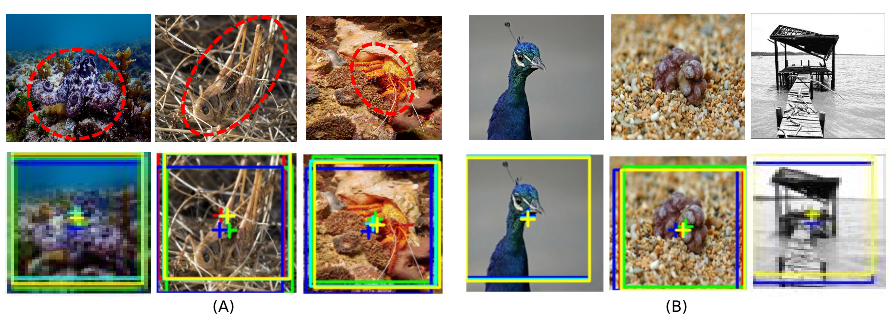

Fig. 7: Ablation study on COD10k camouflaged animal dataset and MSRA10k salient image dataset, for probabilistic method. For each original image in COD10k, the object area is marked. Red bounding box or centroid signifies no channels were excluded, blue means darg excluded keeping all other channels as it is. Green means entropy excluded, cyan signifies Fourier excluded and finally yellow means phot excluded. A comparison with the ground truth of $N = 750$ COD10K images demonstrate that the role of D_{arg} appears context-specific.

8 Energy cost and carbon footprint analysis

The amount of carbon emission calculated according to the formulation outlined in [12]. The model is trained on a PC with an Intel i9-13900K CPU, 64 GB of memory (RAM), and a single NVIDIA GeForce RTX 4080 GPU with 16 GB VRAM. The training loop iterated over 200 epochs with 80%-20% split between training set and testing set. The total power usage in kilowatt-hours (kWh) takes into account training times, mean power drawn during training (P_c) and the power consumed by the GPU (P_g) by accessing the GPU statistics. During training, we also captured the total power drawn from the RAM (P_r). The total power consumption of the training of the model (P_t) is the sum of the power consumed by CPU, GPU and RAM given as

$$P_t = 1.58t(P_c + P_g + P_r)/1000 \tag{3}$$

The CO_2 emission calculation from the total power P_t as in Eq. 3 follows the formulation $CO_2e = 0.954P_t$ along with the other details and assessments as summarized in [12].

9 Image Reconstruction for Cropped Images for Fair Evaluation

To ensure fair comparison between models trained on cropped regions versus full images, we implement a reconstruction-based evaluation methodology that addresses the fundamental challenge of spatial domain mismatch. While training on cropped images (426×426 pixels) offers computational efficiency, direct evaluation against models processing full-resolution images (512×512 pixels) is problematic as the cropped approach reduces the prediction space, potentially inflating metrics without genuine detection improvements. Our solution reconstructs cropped predictions to original image dimensions before evaluation. Given a cropped prediction $\mathbf{P}_c \in [0,1]^{H_c \times W_c}$ extracted from coordinates (x, y, w, h) in the original image space, we reconstruct $\mathbf{P}_r \in [0,1]^{H_o \times W_o}$, shown in equation 4:

$$\mathbf{P}_r(i,j) = \begin{cases} \mathbf{P}_c(i-y, j-x), & \text{if } x \leq j < x+w, \\ & \quad y \leq i < y+h \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

where (x, y) denotes the top-left corner of the crop region, (w, h) its dimensions, and pixels outside the cropped region receive zero probability. We maintain crop metadata for each image in JSON format, storing the filename, crop bounding box coordinates, and original image dimensions:

Our implementation involves: (i) generating predictions on cropped images (426×426 pixels), (ii) initializing zero-filled canvas of original size (512×512), (iii) placing cropped predictions at correct spatial coordinates (x, y) using the stored metadata, and (iv) evaluating reconstructed predictions against original ground truth masks. This ensures evaluation metrics (IoU, precision, recall, F1)

JSON file format to track cropped coordinates for image reconstruction to the original dimension.

```
"crops": [{
  "filename": "image_001.jpg",
  "crop_bbox": {
    "x": 40, "y": 54,
    "width": 426, "height": 426
  },
  "original_image_size": {
    "width": 512, "height": 512
  }
}, ...]
```

reflect performance on the complete image domain, enabling direct comparison with full-resolution models while maintaining computational efficiency during training.

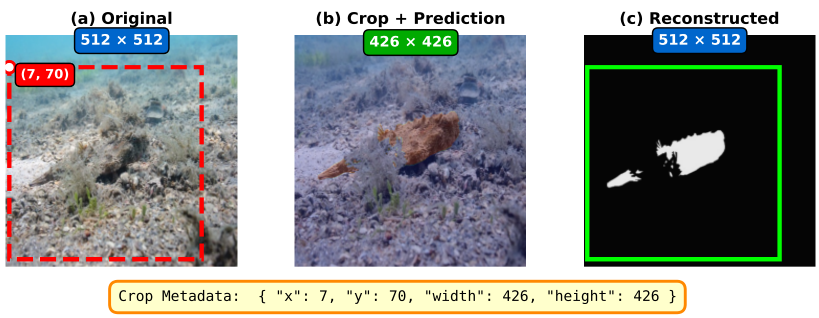

Fig. 8: Image reconstruction process for cropped images

10 CPU-only Architecture Design

Our CPU-only architecture adopts a hierarchical optimization approach that utilizes the hybrid architecture of modern processors. We exploit parallelism at multiple levels: thread, process, and instruction to maximize resource utilization across all available cores while respecting the fundamental memory-bound nature of CPU architectures. Unlike GPUs with their massive parallelism, CPUs excel when data flows efficiently through their cache hierarchy. We implement cache-blocking techniques that partition large tensors into tiles sized for L2 cache (2MB per P-core). For convolution operations, our 256×256 pixel tiles align perfectly with 64-byte cache lines, reducing cache misses and memory stalls. We design an intelligent thread allocation technique for our CPU-only architecture. On hybrid processors, we assign performance cores ($n_{p\text{-}cores}$) to handle the heavy lifting: matrix multiplications and convolutions that demand raw computational power. Each P-core runs two threads to maintain high utilization. Meanwhile, efficiency cores take on supporting roles: computing activation functions, running normalization layers, and coordinating data movement between operations. For uniform architectures lacking this P-core/E-core distinction, we adopt an 80/20 split: roughly 80% of available threads handle primary computation while 20% manage coordination and auxiliary tasks. We always reserve at least 10% of total cores for system processes. During execution, our system continuously monitors CPU utilization and cache performance. When utilization drops below threshold τ_{low} or cache misses spike, the system dynamically adjusts thread allocation. This adaptive behavior, combined with NUMA-aware thread binding, ensures optimal performance throughout the entire training process, from the initial forward passes to the final gradient updates.

Algorithm 6 Adaptive CPU Resource Optimization

Require: CPU topology $\mathcal{T}$, memory configuration $\mathcal{M}$, workload $\mathcal{W}$
Ensure: Optimal thread allocation $(n_{train}, n_{coord}, \phi)$

```
Identify CPU architecture: arch ∈ {uniform, hybrid}
Reserve system threads: n_sys ← max(2, ⌈0.1 · n_logical⌉)
Available threads: n_avail ← n_logical − n_sys
if arch == hybrid then
    n_train ← 2 · n_p-cores {Performance cores}
    n_coord ← min(8, ⌊n_e-cores/2⌋) {Efficiency cores}
else
    n_train ← ⌊0.8 · n_avail⌋
    n_coord ← ⌈0.2 · n_avail⌉
end if
Initialize affinity map: φ ← BindThreads(n_train, n_coord) {NUMA-aware}
while training active do
    u ← GetCpuUtilization()
    if u < τ_low or cache misses > τ_cache then
        AdjustThreads(n_train, n_coord, φ)
    end if
end while
return (n_train, n_coord, φ)
```

11 CPU-Only U-Net Implementation Details

To validate the proposed CPU optimization pipeline, we trained a standard U-Net architecture [10] from scratch. This section details the specific dataset preparation, model architecture, and hardware configuration referenced in Table 2 and Table 3 of the main paper.

Category	Specification
	Hardware Specification
Processor	Intel Core i9-13900K
Memory	64 GB RAM
Core Config	8 P-Cores (Hyper-threaded) + 16 E-Cores
Total Threads	32 Logical Threads
	Training Hyperparameters
Optimizer	AdamW [9]
Learning Rate	1×10^{-4}
Scheduler	ReduceLROnPlateau
Batch Size	12
Patience	15 Epochs (Early Stopping)
Total Epochs	100

Table 3: Hyperparameters and Hardware Configuration for CPU Training.

Dataset and Pre-processing: We experimented with the Kvasir-SEG polyp segmentation [7] and CVC-ColonDB [1] datasets. The combined dataset consists of 2,000 original images and 2,000 corresponding ground truth masks. We adopted an 80-20 split, resulting in 1,600 training and 400 validation samples. While original resolutions varied (ranging from 332×487 to 1920×1072 pixels), all images were resized to 256×256 to optimize training efficiency while maintaining feature integrity. Data augmentation included horizontal/vertical flips, random rotations, and brightness/contrast adjustments.

Model Architecture: We implemented the standard U-Net with a depth of 4 (starting with 64 filters), resulting in approximately 31 million parameters $(31,043,521)$.

- **Input:** 3-channel RGB ($256 \times 256 \times 3$)
- **Output:** 1-channel binary mask ($256 \times 256 \times 1$)
- **Objective Function:** Composite loss of Dice Coefficient + Binary Cross Entropy (BCE).

Hardware and Training Configuration: The training was performed entirely on a CPU to demonstrate the efficacy of the thread-level parallelism strategies. The specific configuration is listed in Table 3.

Performance Results: The model converged after 94 epochs with a total training time of 11 hours, 10 minutes, and 46 seconds. The final model achieved a validation Dice score of 0.8686 with a final validation loss of 0.3002.

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